

Customer Behavior Analysis & Business Decision Making

Minor Project Report

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❖ INTRODUCTION

- This project focuses on analyzing customer behavior using statistical techniques.
- The dataset used is Mall Customer Segmentation dataset.
- Objective is to understand spending patterns and help businesses make decisions.

❖ 2. DATA UNDERSTANDING

- Dataset contains customer demographic and spending data
- Total records: *write from your output*
- No missing values found
 - **Feature Types:**
- Numerical: Age, Income, Spending Score
- Categorical: Gender

❖ 3. DESCRIPTIVE STATISTICS

Table:

Feature Mean Median Std Dev

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- Income shows highest variability
 - Spending score shows moderate variation
 - Customers are diverse in nature
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❖ 4. DATA VISUALIZATION

❖ Insert:

- Histogram images
 - Boxplot image
-

- Age distribution is slightly skewed
 - Income shows wide variation
 - Outliers detected in boxplot
-

❖ 5. GROUP ANALYSIS

- Female customers spend slightly more
 - Difference is small
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❖ 6. CORRELATION ANALYSIS

- Weak correlation between income and spending
 - Negative relation between age and spending
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❖ 7. HYPOTHESIS TESTING

Hypothesis:

- H_0 : No difference in spending
 - H_1 : Difference exists
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Result:

- $p\text{-value} = \text{your output}$
-

Conclusion:

- $p > 0.05 \rightarrow$ Not significant
 - Gender does not affect spending
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❖ 8. CONFIDENCE INTERVAL

- 95% confidence interval calculated
 - Indicates range of average spending
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Result:

- $CI = \text{value whatever we have in output(Notebook)}$
-

❖ 9. BUSINESS INSIGHTS :

- Target high spending customers
 - Focus on young customers
 - Income alone is not a strong factor
 - Gender-based marketing is not necessary
 - Personalized marketing is recommended
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❖ 10. CONCLUSION

- Customer behavior varies significantly
 - Statistical analysis helps in decision making
 - Business should focus on segmentation and personalization
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❖ 11. REFERENCES

- Dataset source (Kaggle / GitHub)
- Python libraries (Pandas, Matplotlib, Scipy)